

Background

Fragile X Syndrome (FXS):

Most common inherited cause of intellectual disability.

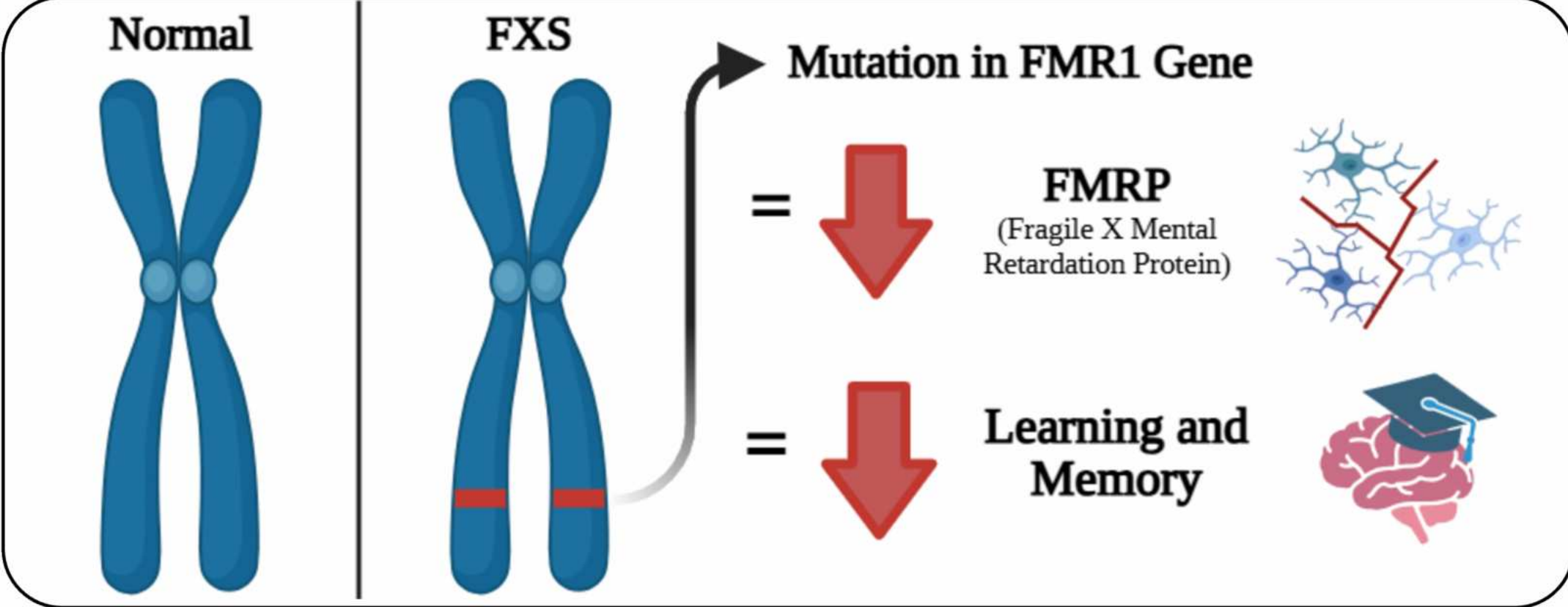


Figure 1: Normal vs. FXS X-chromosome and its effects (Image created by Finalist using BioRender)

Bisphenol A (BPA):

- Chemical used in the production of plastics and epoxy resins.
- Found in plastic water bottles, food containers, canned food, and some medical devices.
- Endocrine Disrupting Chemical (EDC):** interferes with hormones by blocking, mimicking, or disrupting its function, altering neurotransmitter signaling.

Hormones:

Ecdysone

(analogous to human Estrogen)

- Higher levels in females
- Enhanced **synaptic plasticity**
- Strengthens verbal and episodic memory

Juvenile Hormone (JH)

(analogous to human testosterone)

- Higher levels in males
- Supports **spatial navigation**
- Outperformed by females in various memory tasks

Previous research:

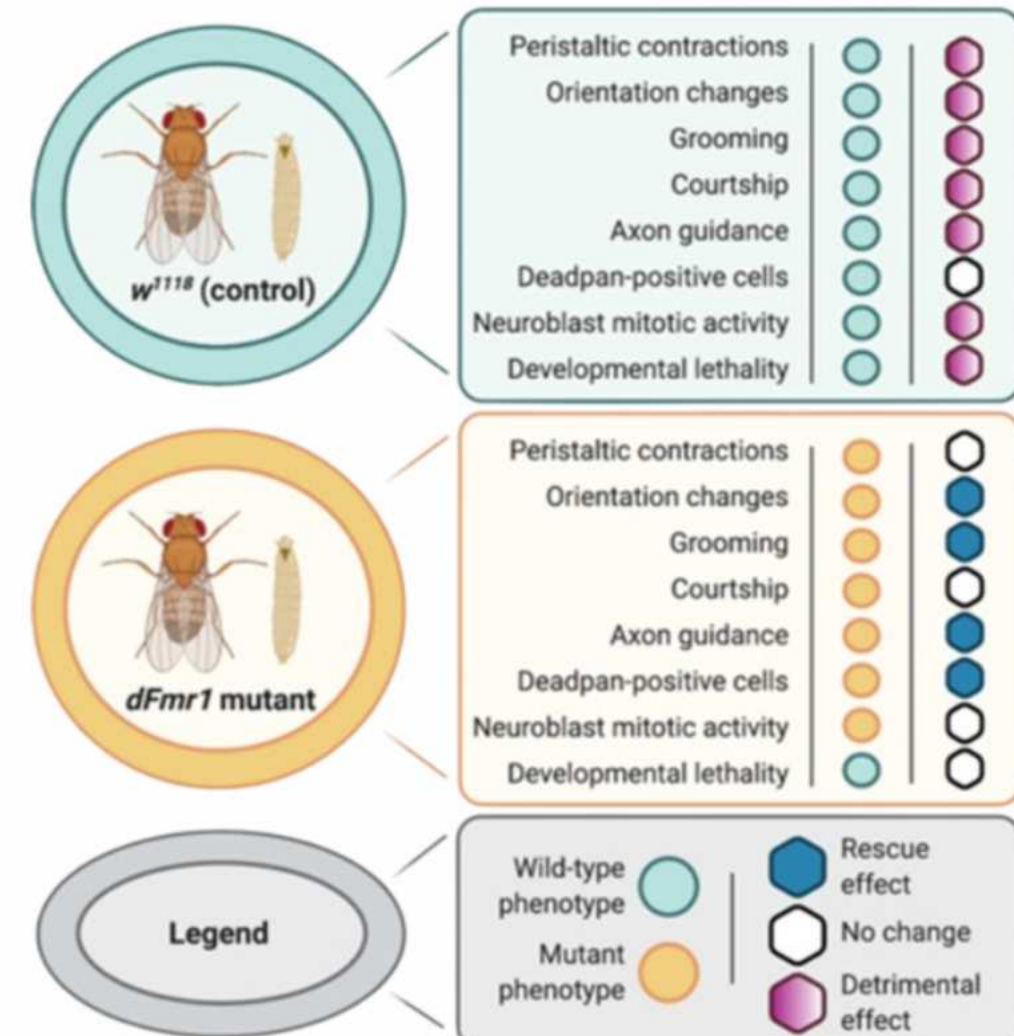


Figure 2: Results of previous research study (Nguyen et al., 2021)

<https://www.sciencedirect.com/science/article/pii/S0168125320305044?via=ihub>

- Behavioral study revealed that BPA had a **rescue effect** on some FXS fly behaviors, but negatively impacted all WT fly behaviors.
- Results were unexpected as BPA should exacerbate the already diminished behaviors of flies with FXS.

Purpose

The research investigates the effect of **exposure to BPA** on the **learning and memory** in male and female *Drosophila melanogaster* with FXS to evaluate whether BPA alters memory regulating hormones differently by gender and in genetically predisposed individuals.

Hypothesis:

When exposed to BPA, FXS male flies would have **greater memory impairment** compared to FXS female flies as the high levels of ecdysone in females would protect against the interfering effect of BPA.

Novelty:

This study addresses the gap in existing research that has focused on male subjects rather than **sex-specific cognitive effects**, while determining the impact of BPA on learning and memory, a **key deficit that has not been studied**.

The Effect of Bisphenol A on the Learning and Memory in Male and Female *Drosophila melanogaster* of the Fragile X Syndrome Model

Variables

Independent Variable:

- Presence of BPA
- Fly Model: w[1118] and FXS
- Gender: Male and Female

Dependent Variable:

- Learning and memory
- Control:**
- WT and FXS groups without BPA

Groups & Expected Results

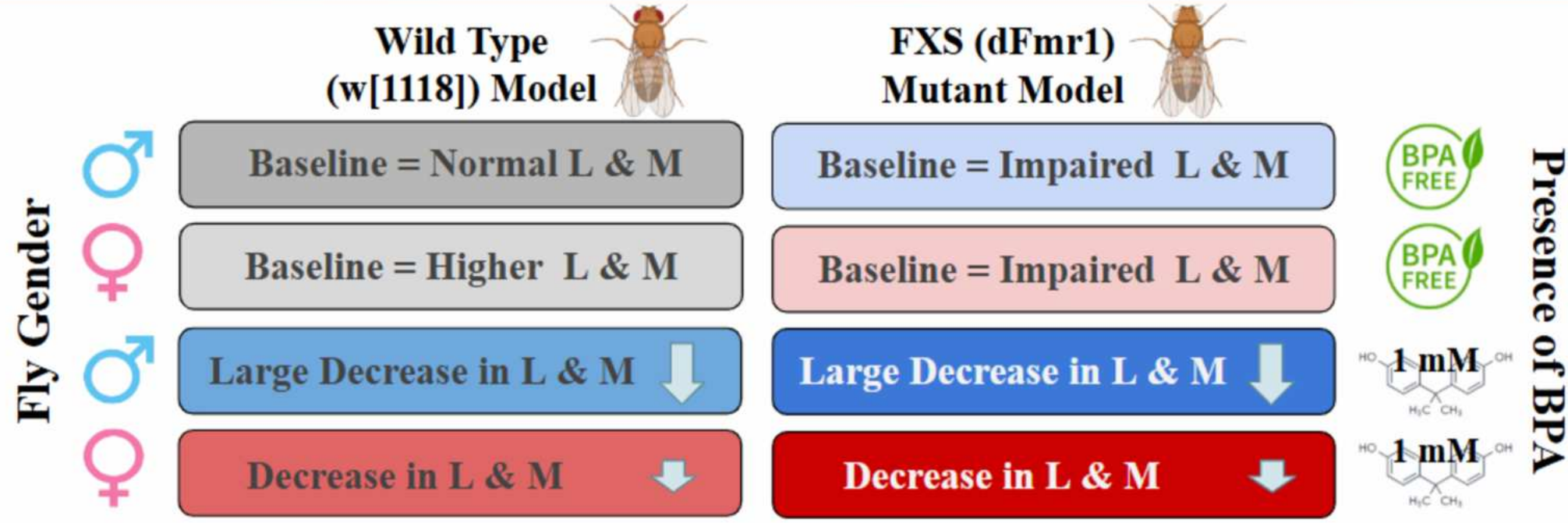


Figure 3: Groups and their expected results; L&M = learning and memory (Image created by Finalist using Canva)

Method

Toxicity Assay:

Last year, a toxicity assay was conducted with the following parameters: 3 concentrations, 10 male flies each, and 2 weeks of exposure.

The least lethal concentration was used for the treatment this year.

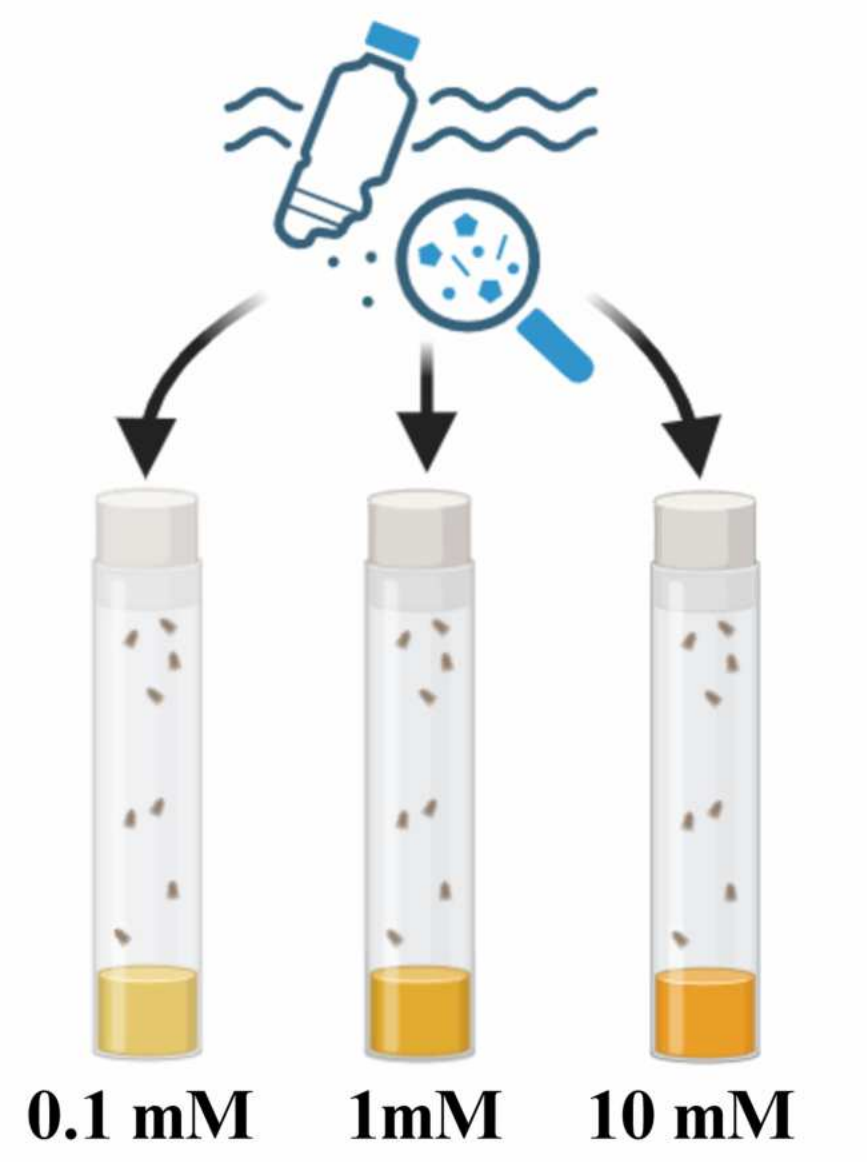


Figure 4: Toxicity assay BPA exposure (Image created by Finalist using BioRender)

Aversive Phototaxis Suppression (APS) Assay:

Trains flies to suppress their natural attraction for light by pairing it with an aversive stimulus, quinine, and later testing in an odorless vial to measure memory and learning.

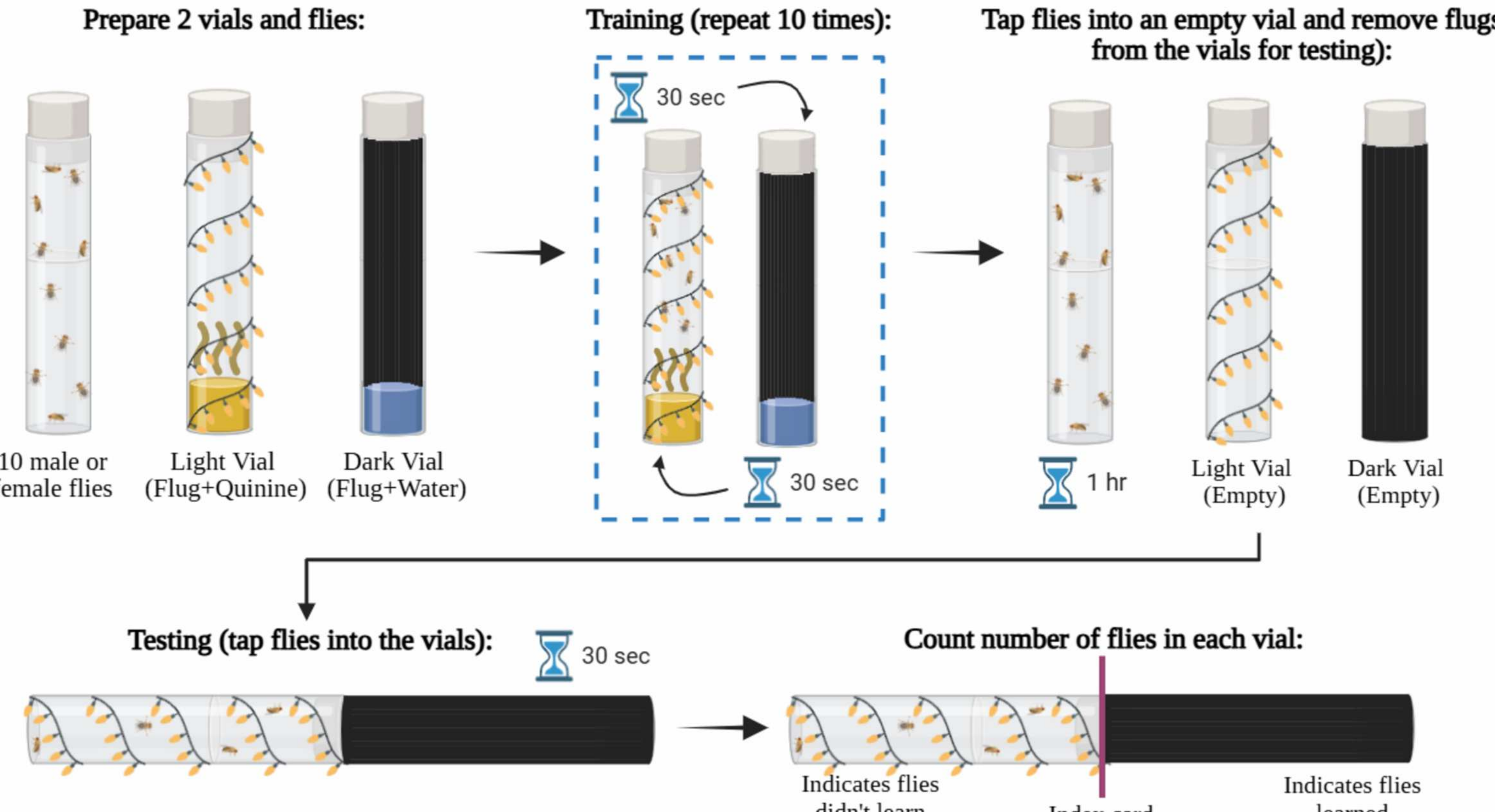


Figure 5: APS Assay setup and procedure (Image created by Finalist using BioRender)

Analysis:

Kruskal-Wallis test with the Dunn's Post-Hoc analysis to evaluate the differences in learning and memory between the eight groups.

APS Assay Results

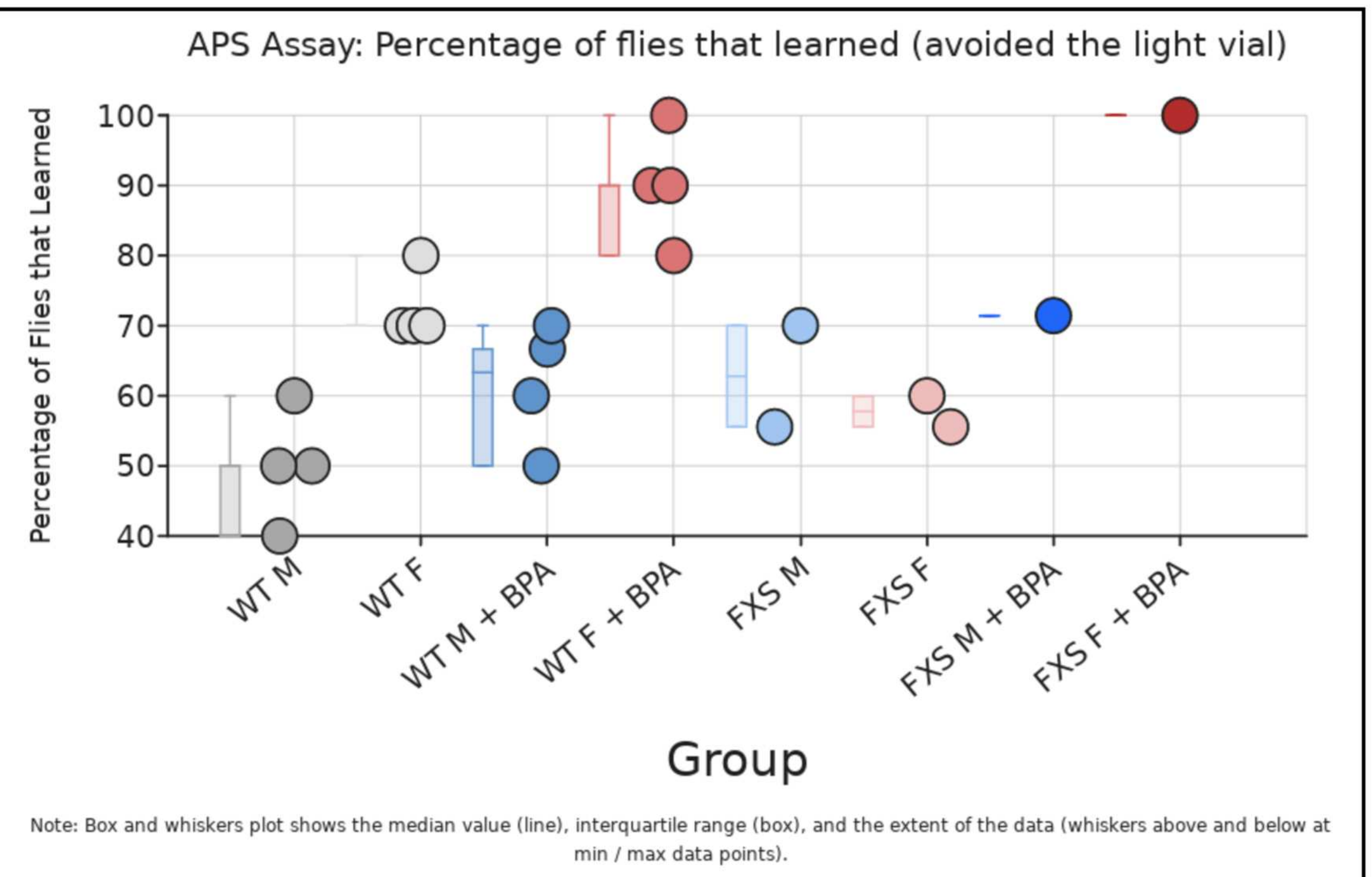


Figure 6: APS Assay graph of the percentage of flies that avoided the light vial (Graph created by Finalist using Data Classroom)

WT = Wild Type | FXS = Fragile X Syndrome
M = Male | F = Female | +BPA = BPA Exposure

Kruskal Wallis Analysis: P-value = 0.01

Dunn's Post-Hoc Analysis:

Pairwise Comparison		P-value
WT M	WT F	0.63
WT M	FXS F + BPA	0.38
WT M + BPA	WT F + BPA	0.44
WT F + BPA	FXS F	0.62

Table 1: APS Assay statistical pairwise comparisons

Toxicity Assay Results

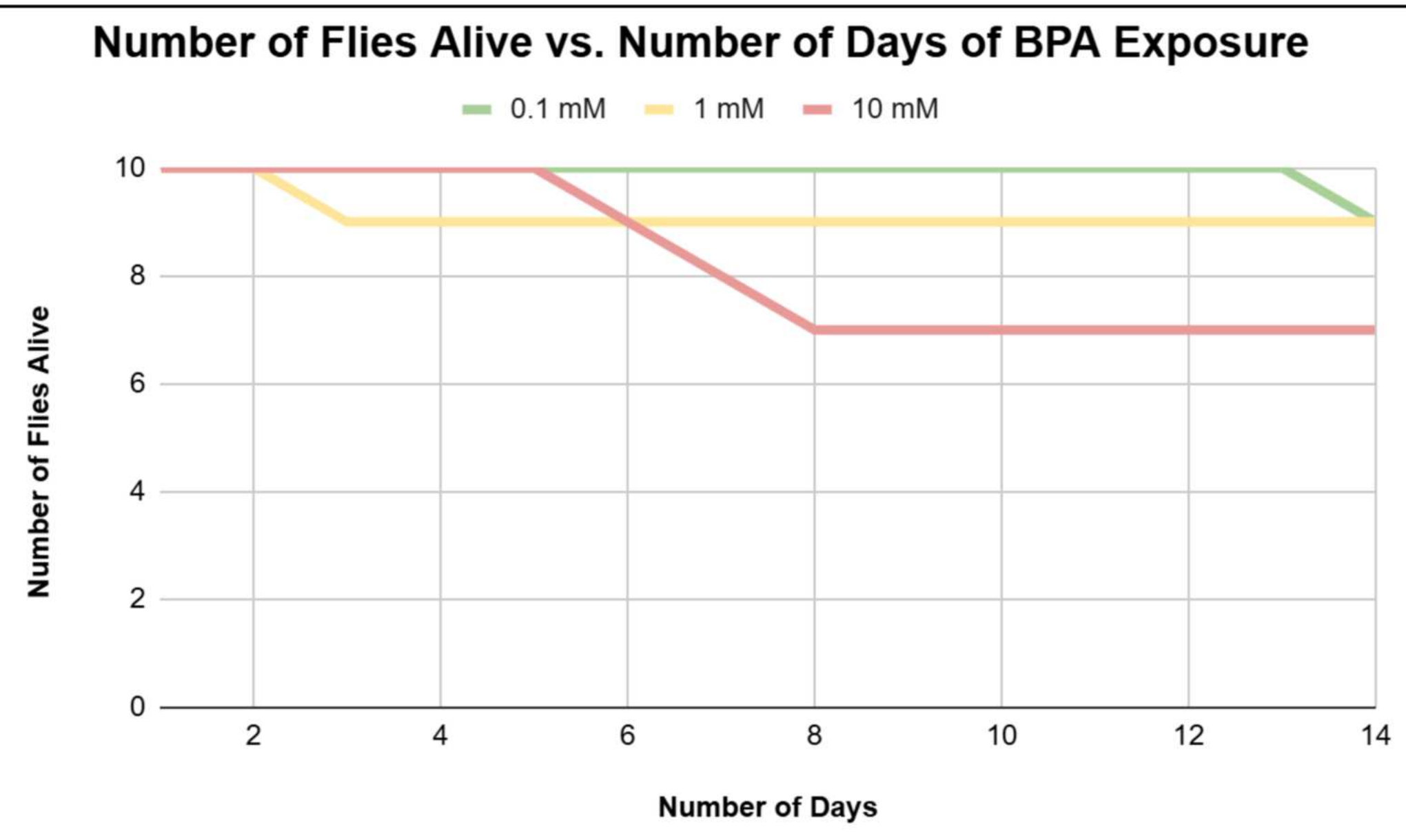


Figure 7: Toxicity assay graph to assess the lethality of the BPA concentrations (Graph created by Finalist using Google Sheets)

Optimal Concentration: 1 mM

Conclusion

- Kruskal-Wallis** test indicated a statistically significant difference among the groups ($p = 0.01$), suggesting that the BPA exposure and biological differences may have influenced memory performance.
- However, the **Dunn's post-hoc** analysis showed none of the pairwise competitions were statistically significant (likely due to small sample sizes and limited data).
- Data collection is still ongoing, but current trends suggest that:

- Females exposed to BPA show **higher memory and learning**.
- Male flies had **roughly consistent performance** despite model and BPA supplementation, which is **unexpected** as it should have a detrimental effect.

So far, the data **does not support my hypothesis** that FXS males exposed to BPA would exhibit greater memory impairment compared to FXS females.

Future Work

- Increasing the number of trials to increase reliability and better evaluate differences in the groups.
- Subsequent studies should **expand BPA concentrations** and further analyzing sex-specific responses.
- Could serve as a **potential treatment** for FXS individuals at minimal dosages if these studies yield positive results.

Citations

Bauer, B. A. (2023, March 24). What is BPA, and what are the concerns about BPA? Mayo Clinic. <https://www.mayoclinic.org/healthy-lifestyle/nutrition-and-healthy-eating/expert-answers/bpa/faq-20058331>

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